

The Mendez–Pinto–Villarreal conjecture in unmixed height three

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Abstract

Let I be a monomial ideal whose minimal primary components are irreducible and whose associated primes all have height three. We prove that equality of all symbolic and ordinary powers forces the exponent of each variable to agree in every primary component containing it. Consequently I is a standard variable weighting of its squarefree radical, which is itself *Simis*. This proves the unmixed height-three case of the Mendez–Pinto–Villarreal conjecture, without restrictions on the exponents or on the supports of minimal generators. The proof uses a classical complete-intersection criterion in dimension one to obtain rigidity on four variables. A putative disagreement then localizes to five variables and four support configurations. Three are excluded by symbolic-power witnesses or triangle specializations; the last admits an explicit obstructing vertex of the irreducible polyhedron for every choice of exponents.

1 Statement and context

Let $S = k[x_1, \dots, x_n]$, where k is an arbitrary field. For an ideal $I \subsetneq S$ its symbolic powers in this paper are

$$I^{(m)} = \bigcap_{\mathfrak{p} \in \text{Min}(S/I)} (I^m S_{\mathfrak{p}} \cap S).$$

The ideal is *Simis* if $I^{(m)} = I^m$ for every integer $m \geq 1$. An irreducible monomial ideal is generated by positive powers of distinct variables. We assume throughout the main theorem that

$$I = \bigcap_{P \in \mathcal{H}} Q_P, \quad Q_P = (x_i^{a_{P,i}} : i \in P), \quad a_{P,i} \geq 1, \quad (1)$$

where $\mathcal{H}$ is a family of distinct three-element subsets of $\{1, \dots, n\}$. Thus (1) is a minimal primary decomposition without embedded primes. Conversely, every monomial ideal with irreducible minimal primary components and all associated primes of height three has this form. This spells out the meaning of minimal irreducible decomposition used here: there is exactly one irreducible primary component for each associated prime.

For positive integers $w_1, \dots, w_n$ and a monomial ideal J , denote by J_w the ideal generated by the images of its monomials under $x_i \mapsto x_i^{w_i}$. This is a standard variable weighting.

Theorem 1. *For an ideal as in (1), the following are equivalent:*

1. *I is *Simis*.*
2. *For each i , the integers $a_{P,i}$ are equal as P ranges over the members of $\mathcal{H}$ containing i , and $\sqrt{I}$ is *Simis*.*
3. *$I = (\sqrt{I})_w$ for a standard variable weighting w , and $\sqrt{I}$ is *Simis*.*

Méndez, Vaz Pinto, and Villarreal [8], Conjecture 5.7, propose the implication from the Simis property to standard weighting for monomial ideals without embedded primes whose irreducible decomposition is minimal. Their Corollary 5.5 already proves invariance of the Simis property under standard weighting; the new implication in Theorem 1 is agreement of the primary-component exponents in unmixed height three.

The height-two case follows from Aslam [1], Theorem 2.3, and is also formulated as a case of the conjecture by Bijender and Kumar [2], Theorem 4.9. Bordoloi, Das, and Kumar [3, 4] treat ideals constrained by the supports of their minimal generators; in particular [4], Theorem 3.10, treats support-three ideals, in the setup of that paper. Component height three and generator support three are different conditions. Our argument makes no assumption on generator support sizes or on their equality with the supports of the radical's generators. The general conjecture remains outside the scope of this paper.

The main mechanism is local: two components containing a discrepantly weighted variable have union of size at most five. Four-variable localizations give a rigid complete-intersection structure. The remaining five-variable configurations can then be excluded uniformly in all positive exponents.

2 Monomial operations and dimension-one rigidity

We first record the operations used in the proof. In this section the sets in $\mathcal{H}$ can have arbitrary sizes, but are distinct and incomparable under inclusion. Write $\mathfrak{p}_P = (x_i : i \in P)$. The power Q_P^m is $\mathfrak{p}_P$ -primary: membership of a monomial depends only on its exponents in P . Localization at $\mathfrak{p}_P$ removes the other components. Consequently

$$I^{(m)} = \bigcap_{P \in \mathcal{H}} Q_P^m. \quad (2)$$

For later use, the exact membership test is

$$x^b \in Q_P^m \iff \sum_{i \in P} \left\lfloor \frac{b_i}{a_{P,i}} \right\rfloor \geq m. \quad (3)$$

Indeed, the right side says that m pure-power factors from Q_P can be chosen to divide x^b .

Lemma 2. *Suppose I is Simis and has the decomposition just described.*

1. *For a set T of variables, the ideal in $k[x_i : i \in T]$ obtained by retaining exactly the components with $P \subseteq T$ is Simis, if any components remain. Their exponents are unchanged.*
2. *Let $\pi : S \longrightarrow k[x_i : i \in T]$ set all other variables equal to zero. Suppose every $\pi(Q_P)$ is proper and nonzero, and their radicals are distinct and incomparable after deleting identical duplicate ideals. Then their intersection $\pi(I)$ is Simis.*

Proof. For (1), invert the variables outside T . A component disappears precisely when its support is not contained in T . Localization commutes with powers and finite intersections, so the Simis equalities (2) pass to the retained components. They descend to $k[x_i : i \in T]$ by faithful flatness of Laurent polynomial extension.

For (2), if L is monomial, a monomial in the remaining variables belongs to $\pi(L)$ exactly when it belongs to L . Hence π commutes with intersections of monomial ideals; as a surjective homomorphism it also commutes with ideal powers. Apply π to $I^m = \bigcap_P Q_P^m$. Under the stated hypothesis the resulting components form a minimal irreducible primary decomposition, so (2) identifies the right side with $\pi(I)^{(m)}$. $\square$

Part (2) is deliberately conditional. Setting variables equal to zero is not asserted to preserve the Simis property without checking the image decomposition. In both applications below that check is explicit.

We use the following classical local criterion in its ideal case: a generically complete intersection ideal of grade at least two in a Cohen–Macaulay local ring is a complete intersection if all its ordinary-power quotients are Cohen–Macaulay. This is the rank-one specialization of Branco Correia–Zarzuela [5], Theorem 6.4. Its mechanism is Burch’s inequality, which forces equimultiplicity, followed by the fact that a generically complete intersection equimultiple ideal is a complete intersection. The latter criterion and its proof also appear in [6], Theorem 5.3.

Lemma 3. *Let $h \geq 2$. Suppose $K \subset k[z_1, \dots, z_{h+1}]$ is Simis and its minimal primary components are irreducible of height h . Then K is a monomial complete intersection, has at most two minimal primes, and the exponents of any variable agree in all components containing it.*

Proof. Localize at the homogeneous maximal ideal $\mathfrak{m}$. For every $m \geq 1$, the equality of symbolic and ordinary powers expresses K^m as an intersection of powers of its primary components. Those powers are primary with height- h radicals. There are no embedded associated primes, so the one-dimensional local quotient by K^m has depth one and is Cohen–Macaulay. At every minimal prime, K is generated by the h pure powers in its corresponding irreducible component. It is therefore generically a complete intersection. The local ring is regular and K has grade $h \geq 2$, so the preceding classical criterion applies.

The number of minimal monomial generators is unchanged on localizing at $\mathfrak{m}$, as is seen from $K/\mathfrak{m}K$. Thus K has exactly h minimal monomial generators. Their supports are pairwise disjoint. In fact, if two shared a variable, that variable together with one variable from each of the other $h - 2$ generators would give a prime of height at most $h - 1$ containing K , a contradiction. These h disjoint nonempty supports occupy at most $h + 1$ variables. All therefore have size one, with at most one exception of size two. The minimal primes select one variable from each generator support, giving at most two choices.

At such a prime, all unselected variables are units. The localized ideal is generated by the powers of the selected variables occurring in their unique monomial generators. Contraction gives the unique isolated primary component. In particular the exponent of a selected variable is independent of the prime in which it is selected. $\square$

Corollary 4. *An intersection of three irreducible primary ideals with radicals (x, y) , (x, z) , and (y, z) in $k[x, y, z]$ is not Simis.*

Proof. Lemma 3 with $h = 2$ permits at most two minimal primes. $\square$

3 An obstruction on five variables

We give a uniform obstruction needed for the last support configuration. For a monomial ideal L let $\text{NP}(L)$ be its Newton polyhedron, the convex hull of its exponent set. Equivalently it is the convex hull of its minimal generator exponents plus $\mathbb{R}_{\geq 0}^n$. For a decomposition into irreducible components, define

$$\text{IP}(L) = \bigcap_P \text{NP}(Q_P) = \left\{ b \in \mathbb{R}_{\geq 0}^n : \sum_{i \in P} \frac{b_i}{a_{P,i}} \geq 1 \text{ for every } P \right\}.$$

These irreducible polyhedra and their relation to integral closures are studied in Grisalde–Secoleanu–Villarreal [7], Section 7. Only the following necessary condition is required here.

Lemma 5. *Under the decomposition assumptions of Lemma 2, if L is Simis, then $\text{NP}(L) = \text{IP}(L)$. Every vertex of $\text{IP}(L)$ is then the exponent vector of a monomial in L .*

Proof. The inclusion $\text{NP}(L) \subseteq \text{IP}(L)$ follows from $L \subseteq Q_P$. Take a rational $b \in \text{IP}(L)$ and a positive integer N such that every $Nb_i/a_{P,i}$ is integral for every relevant pair (P, i) , and every Nb_i is integral. Formula (3) gives $x^{Nb} \in \bigcap_P Q_P^N = L^N$. Since $\text{NP}(L^N) = N \text{NP}(L)$, we obtain $b \in \text{NP}(L)$. Rational points are dense in the rational polyhedron $\text{IP}(L)$, and $\text{NP}(L)$ is closed, which proves equality.

Finally, every vertex of the convex hull of finitely many generator exponents plus the nonnegative orthant is one of those exponents. Indeed a nonzero orthant displacement places the point strictly between its undisplaced point and twice that displacement, while a nontrivial convex combination contradicts extremality. $\square$

Proposition 6. *For arbitrary positive integers $a, b, c, \alpha, \beta, \gamma, d, e$, the ideal in $k[x, y, z, u, v]$ with components*

$$\begin{aligned} Q_0 &= (x^a, y^b, z^c), & Q_1 &= (x^\alpha, u^d, v^e), \\ Q_2 &= (y^\beta, u^d, v^e), & Q_3 &= (z^\gamma, u^d, v^e) \end{aligned}$$

is not Simis.

Proof. The four radicals are distinct incomparable triples. Use coordinates (X, Y, Z, U, V) and put $r = \alpha/a + \beta/b + \gamma/c$. The irreducible polyhedron has nonnegative coordinates and inequalities

$$\frac{X}{a} + \frac{Y}{b} + \frac{Z}{c} \geq 1, \quad \frac{X}{\alpha} + \frac{U}{d} + \frac{V}{e} \geq 1, \quad \frac{Y}{\beta} + \frac{U}{d} + \frac{V}{e} \geq 1, \quad \frac{Z}{\gamma} + \frac{U}{d} + \frac{V}{e} \geq 1. \quad (4)$$

If $r > 1$, take

$$p = \left(\frac{\alpha}{r}, \frac{\beta}{r}, \frac{\gamma}{r}, d(1 - 1/r), 0 \right).$$

This is feasible. The four equalities in (4) together with $V = 0$ uniquely determine p : the last three give $X/\alpha = Y/\beta = Z/\gamma = t$, the first gives $rt = 1$, and then $U = d(1 - t)$. Hence p is a vertex. Each of $X/a, Y/b, Z/c$ is strictly between zero and one, since its numerator is a positive summand of r and the other two summands are positive.

If $r \leq 1$, instead take

$$p = (a(1 - \beta/b - \gamma/c), \beta, \gamma, 0, 0).$$

Here $X \geq \alpha$, so the point is feasible. The first, third, and fourth equalities in (4), together with $U = V = 0$, uniquely determine it. It is again a vertex. Since $\alpha/a > 0$ and $r \leq 1$, we have $0 < X < a$, $0 < \beta < b$, and $0 < \gamma < c$.

In both cases the vertex has all three first coordinates below the respective pure-power thresholds of Q_0 . It is outside the union of the exponent orthants of Q_0 , hence cannot be the exponent vector of a monomial in $\bigcap_{j=0}^3 Q_j$. Lemma 5 excludes the Simis property. $\square$

4 Proof of the height-three theorem

Assume I is Simis. If two component supports P, R intersect in two variables, their union has size four. Retain the components contained in that union by Lemma 2(1), and apply Lemma 3 with $h = 3$. Their weights agree on both common variables. We will use this conclusion repeatedly.

Suppose some variable nevertheless has different exponents in two components. Those supports must intersect in exactly that variable. After relabeling, and localizing to their union, write

$$P = \{x, a, b\}, \quad R = \{x, c, d\},$$

with unequal x -exponents A and B . Here x, a, b, c, d denote variables; exponent letters in Proposition 6 will only be used after a separate relabeling. Every additional support containing x would have the form $\{x, s, t\}$ with $s \in \{a, b\}$ and $t \in \{c, d\}$. Its two-variable overlaps with P and R would force its x -exponent to equal both A and B . Thus P and R are the only supports containing x .

Call the other supports *extras*. They are triples in $\{a, b, c, d\}$. There are at most two: if there are any, invert x and apply Lemma 3 to the retained height-three components in four variables. Up to exchanging the pairs $\{a, b\}, \{c, d\}$ and permuting within each pair, there are four possibilities.

No extras

Exchange P, R if necessary so that $A < B$, and let C be the c -exponent in Q_R . The monomial

$$f = x^{\max(2A, B)} c^C$$

lies in $Q_P^2 \cap Q_R^2$. In the subring $k[x, c]$ we have

$$I \cap k[x, c] = (x^A) \cap (x^B, c^C) = (x^B, x^A c^C).$$

A factor of a monomial involving only x, c also involves only x, c , so ordinary powers commute with this monomial contraction. The three generator products in degree two have exponent pairs $(2B, 0)$, $(A + B, C)$, and $(2A, 2C)$. None divides f , because

$$\max(2A, B) < 2B, \quad \max(2A, B) < A + B, \quad C < 2C.$$

Therefore $f \in I^{(2)} \setminus I^2$, a contradiction.

One extra

By symmetry the extra has support $\{b, c, d\}$. Set $a = d = 0$. The three image components have supports $\{x, b\}$, $\{x, c\}$, and $\{b, c\}$, respectively. Their radicals are distinct and incomparable. Lemma 2(2) would make their intersection Simis, contrary to Corollary 4.

Two extras omitting variables from different pairs

The extras can be written $\{b, c, d\}$ and $\{a, b, d\}$. They share the two variables b, d , so the four-variable conclusion at the start of the proof makes their exponents on b, d equal. Set $a = c = 0$. The image supports are

$$\{x, b\}, \quad \{x, d\}, \quad \{b, d\}, \quad \{b, d\}.$$

The last two ideals are identical, including their exponents. After deleting one duplicate, Lemma 2(2) again gives a Simis triangle, contradicting Corollary 4.

Two extras omitting variables from the same pair

The extras can be written $\{a, c, d\}$ and $\{b, c, d\}$. Together with R , they all contain c, d . Pairwise four-variable localization makes the exponents on c, d agree across all three. After relabeling the variables (x, a, b, c, d) as (x, y, z, u, v) , the four components have exactly the form of Proposition 6. That proposition supplies the final contradiction.

These possibilities exhaust the subsets of the four available extra triples of cardinality at most two. Thus every shared-variable exponent agrees. Let w_i be this common exponent, choosing $w_i = 1$ for

any variable occurring in no component. Weighting commutes with monomial intersections, by the least-common-multiple formula for their generators. Hence

$$I = \bigcap_{P \in \mathcal{H}} (x_i^{w_i} : i \in P) = \left(\bigcap_{P \in \mathcal{H}} (x_i : i \in P) \right)_w = (\sqrt{I})_w.$$

Corollary 5.5(b) of [8] now makes $\sqrt{I}$ Simis. Conversely the same established weighting invariance proves that every standard weighting of a squarefree Simis ideal is Simis. Uniqueness of isolated primary components identifies the component exponents of such a weighting, so the three conditions of Theorem 1 are equivalent. $\square$

5 Scope and further questions

The theorem removes the nonsquarefree weighting question in the entire unmixed height-three class under the irreducibility hypothesis. It leaves the separate squarefree Simis classification intact. It also does not assert that equality of just the second powers forces the conclusion: the dimension-one criterion uses all powers.

There is a general localization principle behind the argument. If all component heights are at most h , a discrepancy on a shared variable is retained by localizing to the union of the two relevant supports, which has at most $2h - 1$ variables. Thus a counterexample to weighting rigidity in bounded height would already occur in that many variables. In unmixed height three the four-variable complete-intersection lemma makes the remaining support list short. For mixed heights at most three, or unmixed height four, additional configurations remain to be controlled. They are natural next cases; the present proof does not settle them.

The accompanying Python script checks the finite support reduction on five labeled variables and the two rational-vertex formulas using exact arithmetic. These calculations are supplementary: the proof above applies to all positive exponents without a finite cutoff. The manuscript and checks were prepared with OpenAI Codex. The originating system performed the internal audit; no independent human verification or peer review is represented.

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